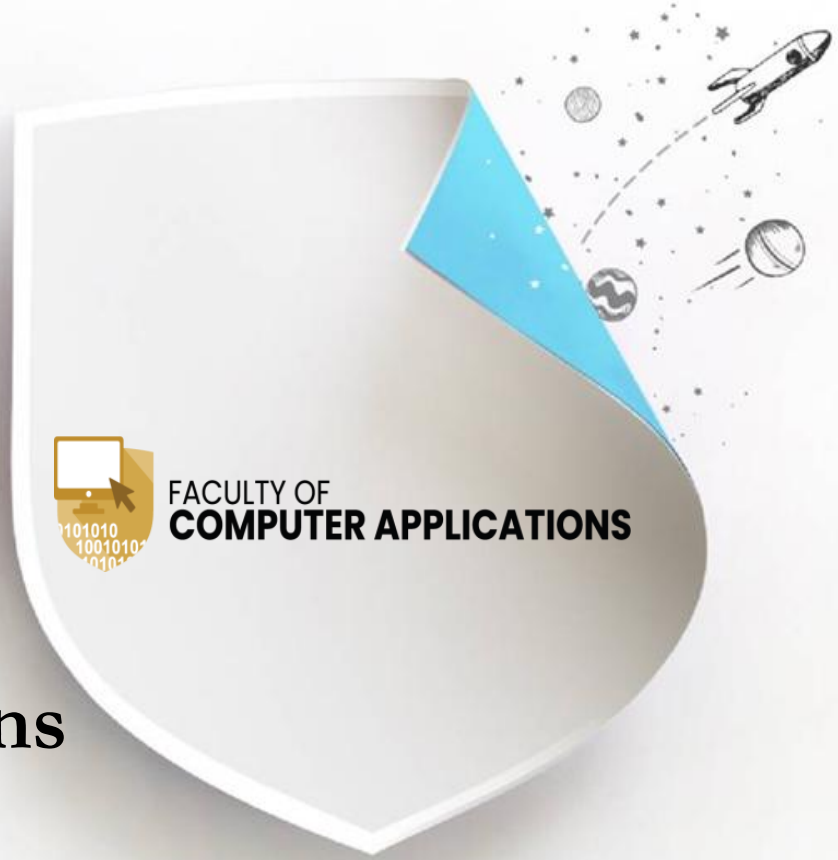
Marwadi University
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NAAC A+



FACULTY OF COMPUTER APPLICATIONS

Bachelor of Computer Applications

BCA Semester: 2

05BC3202 – Analysis and Design of Systems

PROGRAM	**BACHELOR OF COMPUTER APPLICATIONS**
SEMESTER	**2**
COURSE TITLE	**ANALYSIS AND DESIGN OF SYSTEMS**
COURSE CODE	**05BC3202**
COURSE CREDITS	**4**
COURSE DURATION	**20 HOURS**

UNIT 3 – PROCESS OF ANALYSIS

TOPICS COVERED

USING DATA FLOW DIAGRAMS :

- Data flow approach
- Develop data flow diagrams
- Logical and physical data flow diagrams
- Examples

TOPICS COVERED

DATA DICTIONARIES :

- Data dictionary
- Data repository
- Creating a data dictionary
- Using a data dictionary
- Process specifications and structured decisions
- Overview of process specifications
- Structured English
- Decision tables
- Decision trees

DFD [DATA FLOW DIAGRAM]

- DFD means Data Flow Diagram.
- Data flow diagram is a graphical representation of the flow of data in an information system.
- The flow of data of a system or a process is represented by DFD. It also gives insight into the inputs and outputs of each entity and the process itself.
- Data Flow diagrams are very popular because they help us to visualize the major steps and data involved in software-system processes.
- DFD is capable of displaying incoming data flow, outgoing data flow, and stored data.
- DFDs display the flow of data in the system at various levels. DFD does not contain any control or branch elements.

TYPES OF SYSTEM

The main key difference is that Logical DFD focuses on business-related activities while Physical DFD focuses on how the system is implemented.

1. Logical DFD:

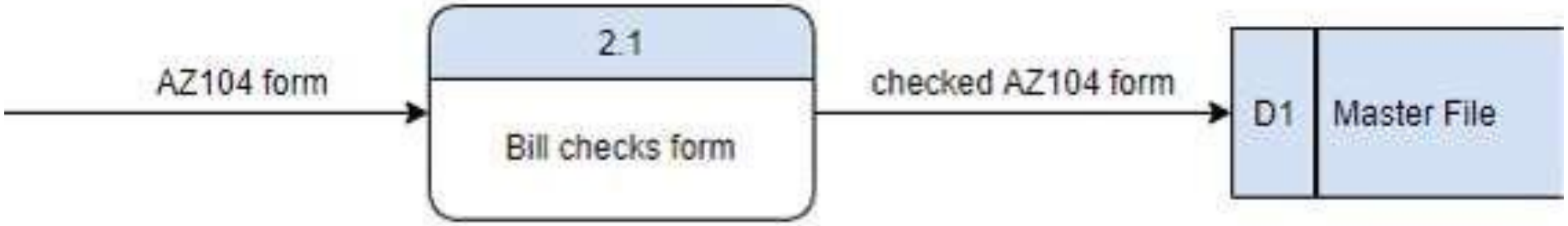
This type of DFD concentrates on the system process, and flow of data in the system. A logical data flow diagram focuses on the business and how the business operates. It is not concerned with how the system will be constructed.

2. Physical DFD:

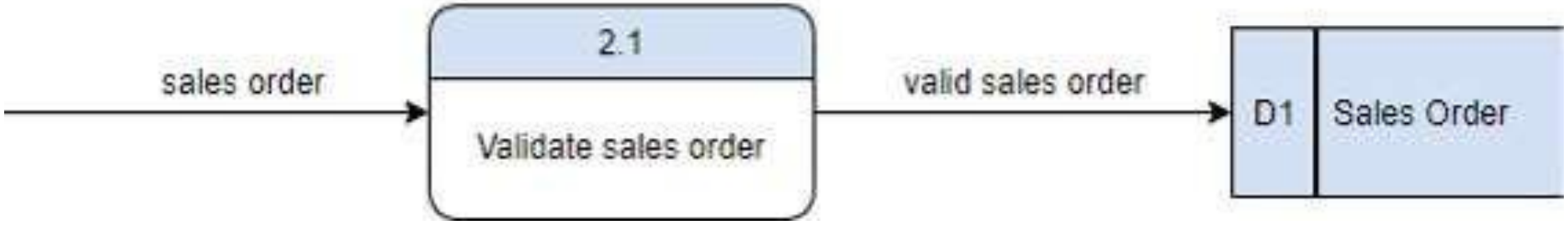
It shows how the data flow is implemented in the system including the hardware, software, files, and people involved in the system. It is more specific and close to the implementation

TYPES OF SYSTEM

Physical DFD



Logical DFD



DFD COMPONENTS

The Data Flow Diagram has 4 components:

1. Entities:

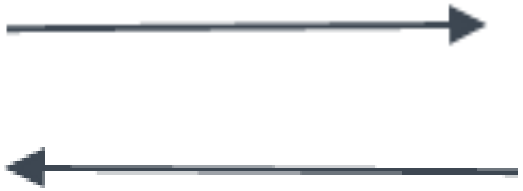
An outside system that sends or receives data, communicating with the system being diagrammed. They are the sources and destinations of information entering or leaving the system. They might be an outside organization or person, a computer system or a business system.



DFD COMPONENTS

2. Data Flow:

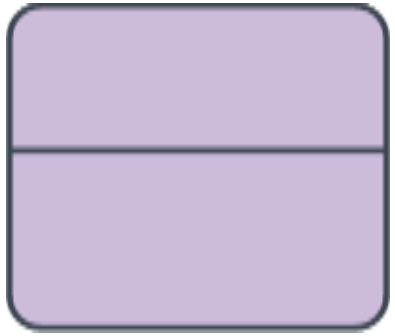
Data flow describes the information transferring between different parts of the systems. The arrow symbol is the symbol of data flow.



DFD COMPONENTS

3. Process:

Activities and actions taken on the data are represented by Circle or Round-edged rectangles.



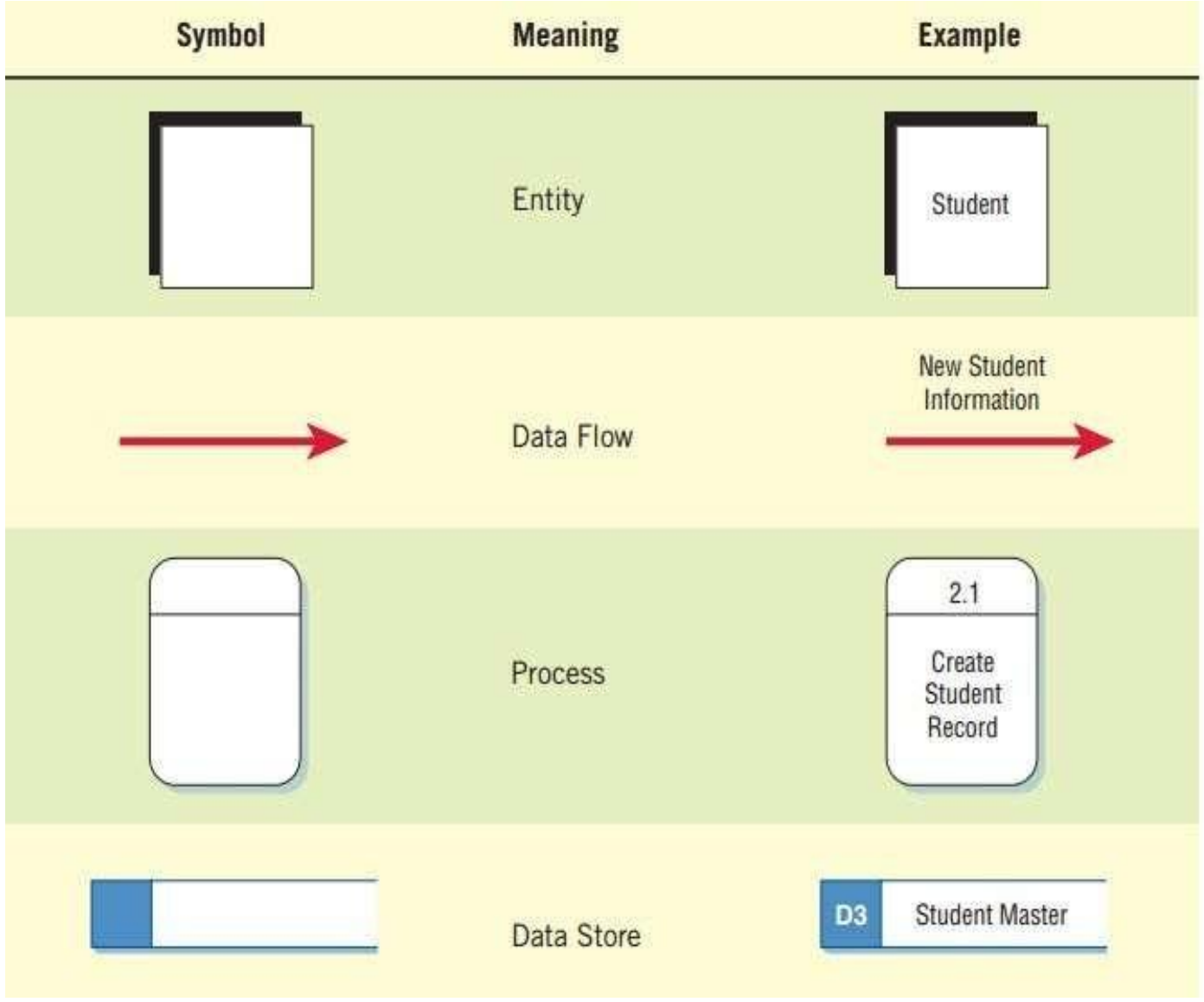
DFD COMPONENTS

4. Data Stores:

places where information is maintained or held, most often databases or database tables.



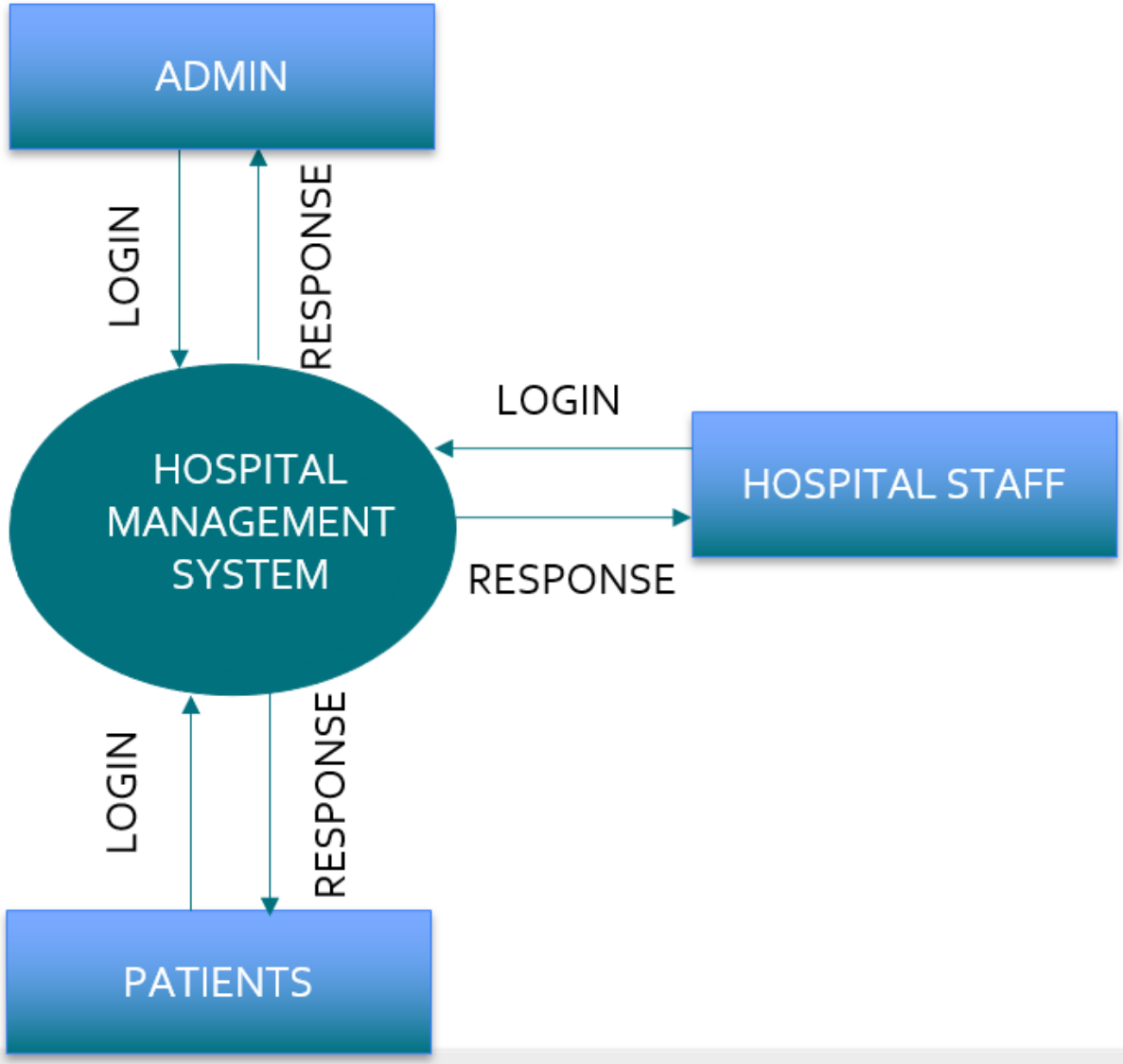
DFD COMPONENTS



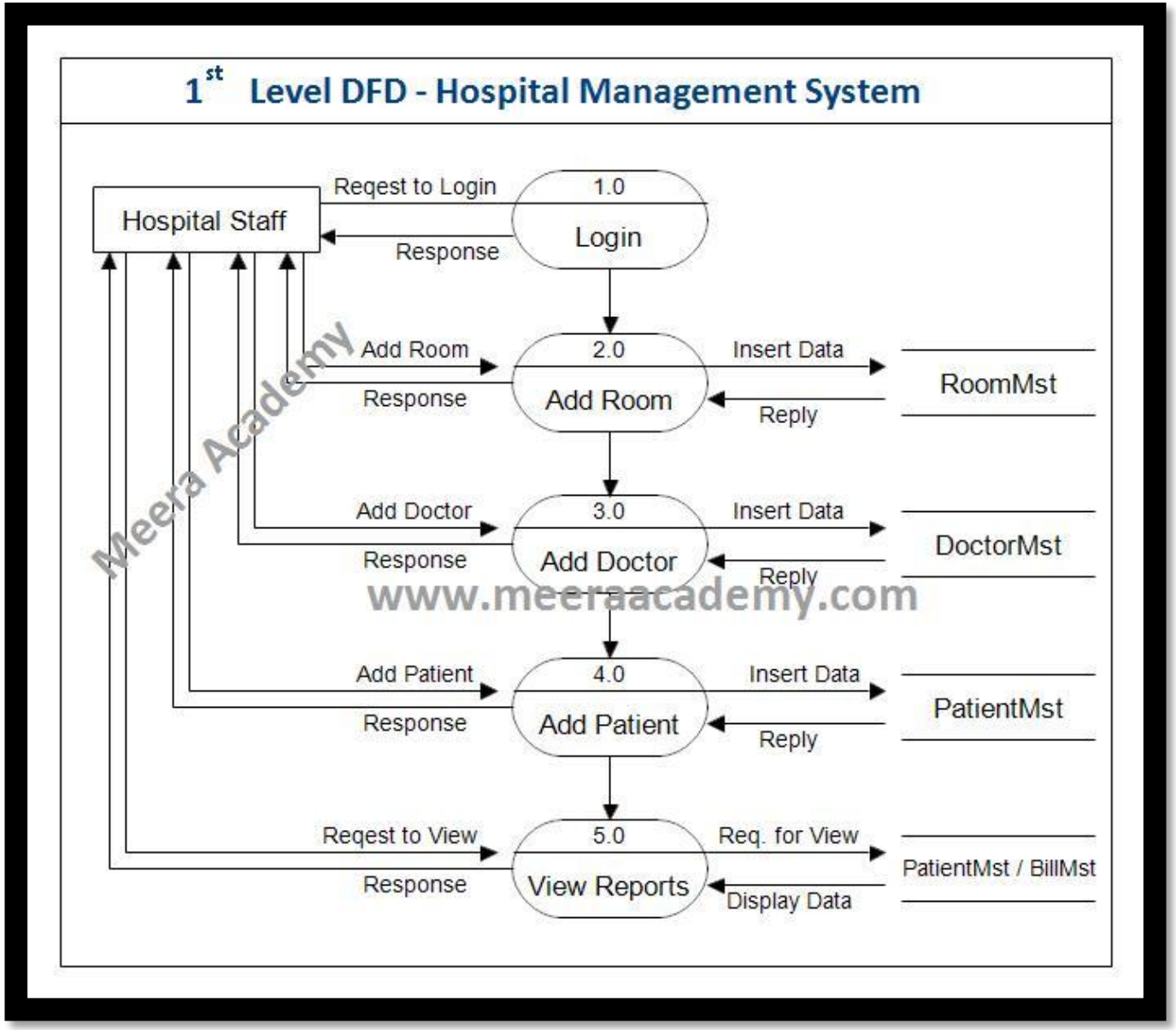
LEVELS OF DFD

- **Level 0:** Also known as a "context diagram", this is the highest level and represents a very simple, top-level view of the system being represented.
- **Level 1:** Still a relatively broad view of the system, but incorporates subprocesses and more detail.
- **Level 2:** Provides even more detail and continues to break down sub-processes as needed.

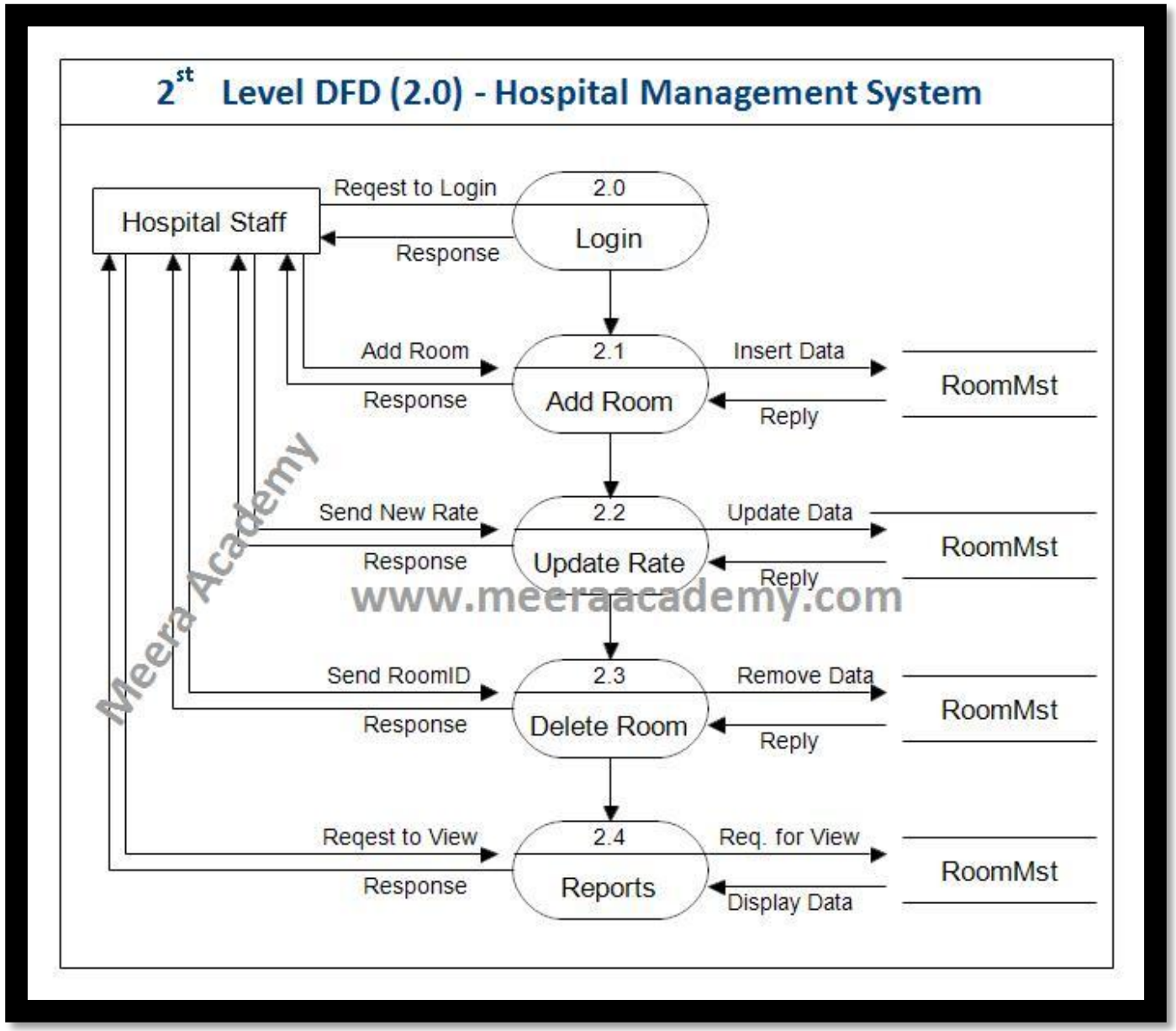
EXAMPLE - LEVEL 0



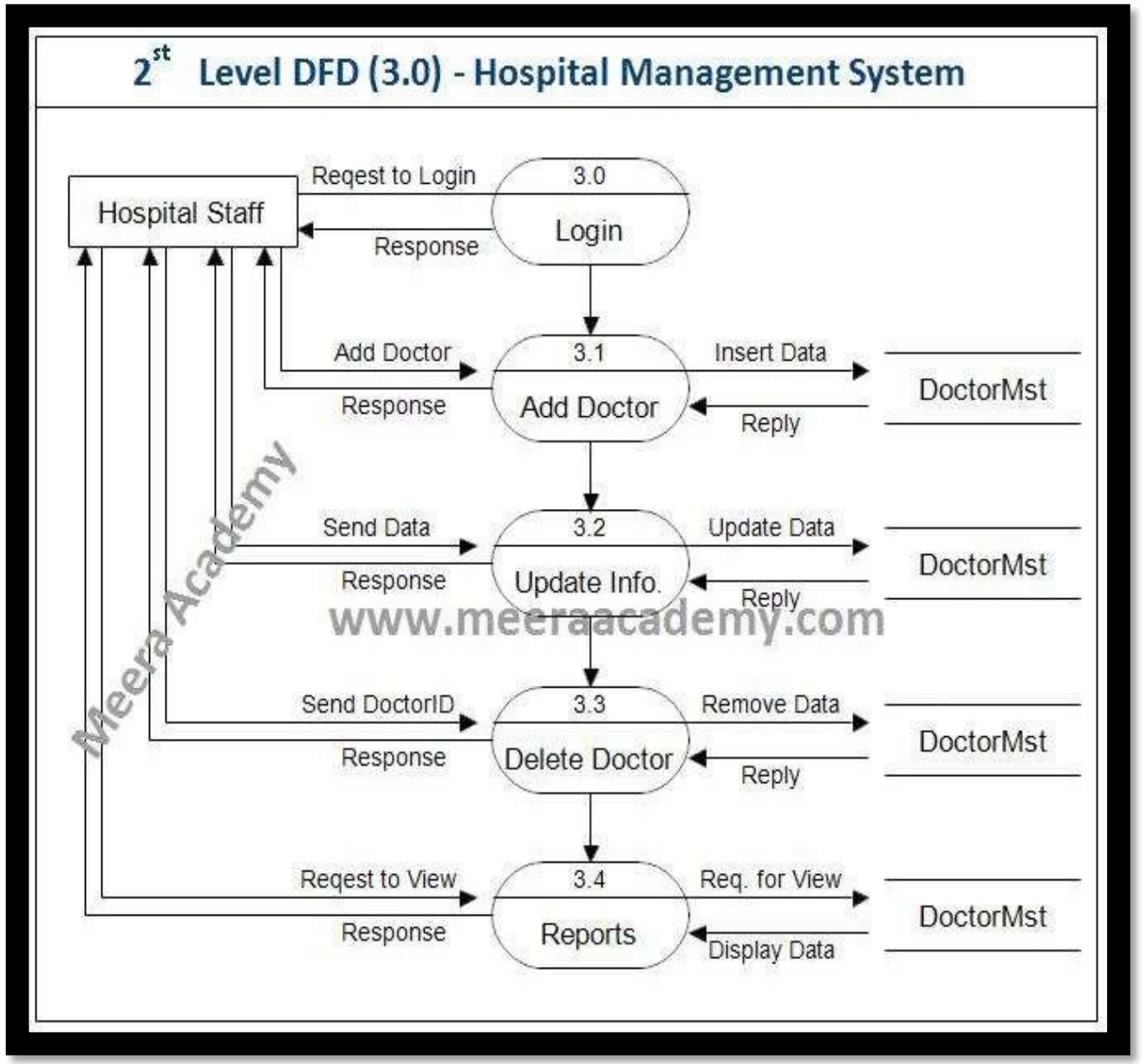
EXAMPLE - LEVEL 1



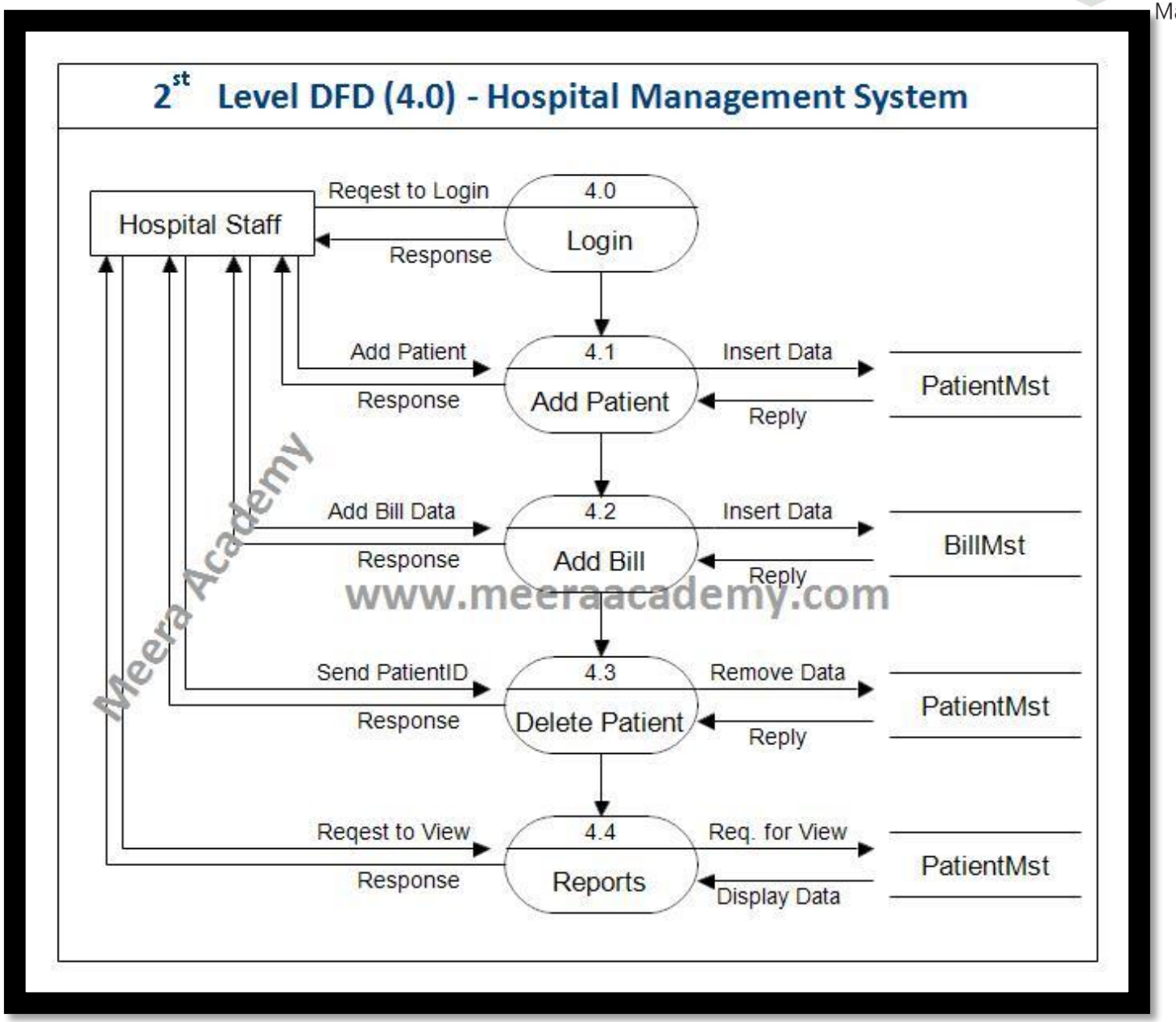
EXAMPLE - LEVEL 2 [2.0]



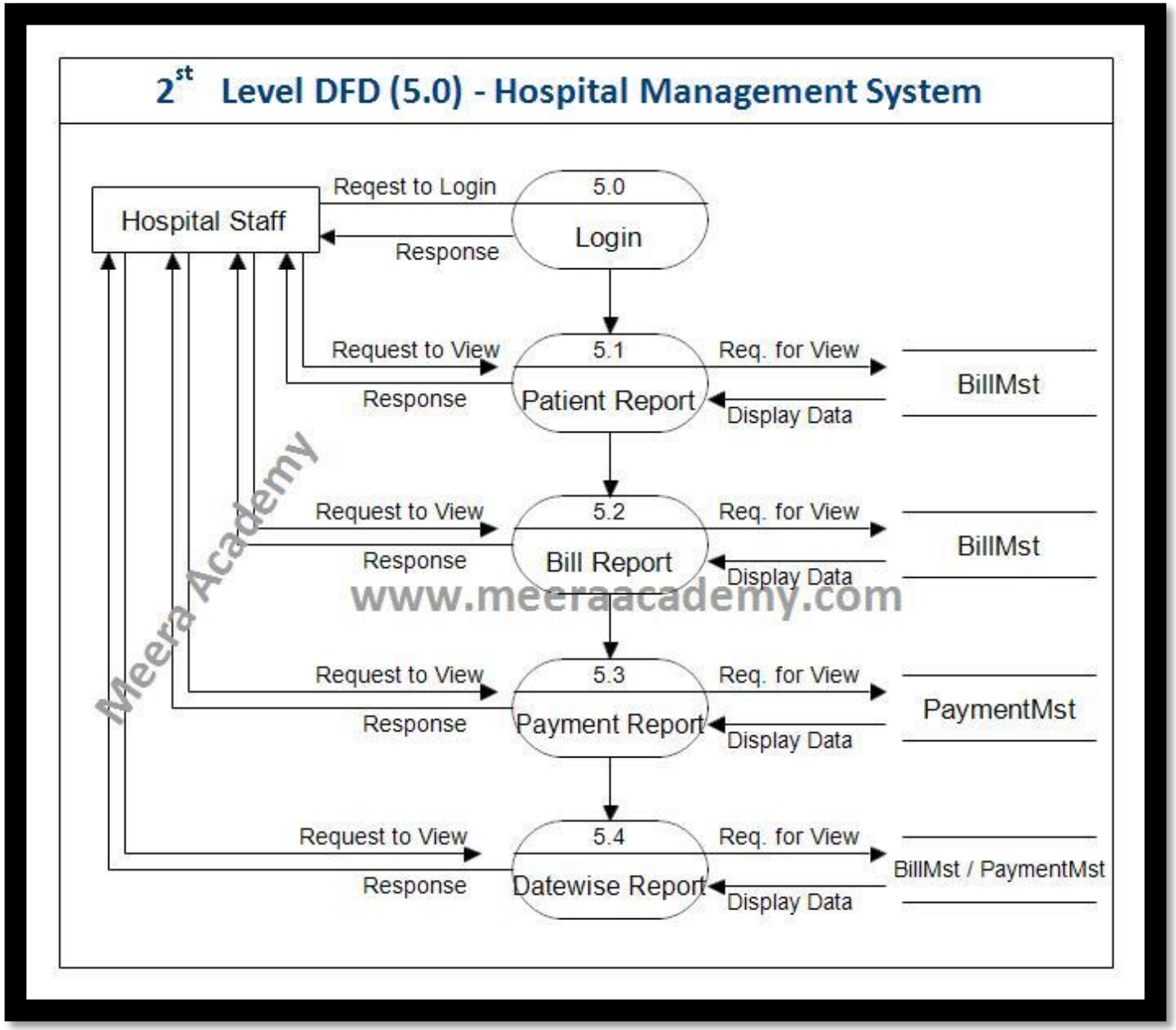
EXAMPLE - LEVEL 2 [3.0]



EXAMPLE - LEVEL 2 [4.0]



EXAMPLE - LEVEL 2 [5.0]



DFD RULES

- The name of the entity should be easy and understandable without any extra assistance (like comments).
- The processes should be numbered or put in ordered list to be referred easily.
- The DFD should maintain consistency across all the DFD levels.
- A single DFD can have maximum processes up to 9 and minimum 3 processes.

DFD ADVANTAGES

- It helps us to understand the functioning and the limits of a system.
- It is a graphical representation which is very easy to understand as it helps visualize contents.
- Data Flow Diagram represent detailed and well explained diagram of system components.
- It is used as the part of system documentation file.
- Data Flow Diagrams can be understood by both technical or nontechnical person because they are very easy to understand.

CASE STUDIES OF DFD

Draw mentioned DFDs on :

1. Library Management System
2. Banking Management System
3. Online Shopping System
4. Online JOB portal

DATA REPOSITORY

- Data repository contains information about data and procedures; a larger collection of project information is called a repository. It includes information about the data maintained in the system, including data, flows, data stores, entities, and messages, diagrams, Screen and report design, Data relationships, Project requirements, Project management information, such as delivery schedules, achievements, issues that need resolving, and project users.

DATA DICTIONARY

- Data dictionary contains information such as what is in the database, who is allowed to access it, where is the database stored, etc.
- It plays an important role in building a database.
- A data dictionary is often referenced as meta-data(data about data.)

Data Dictionary contains:

1. Name of all tables in the database.
2. Names of each field in the tables of the database.
3. Constraints defined on tables.
4. Physical information about tables like their storage location etc.

DATA DICTIONARY EXAMPLE

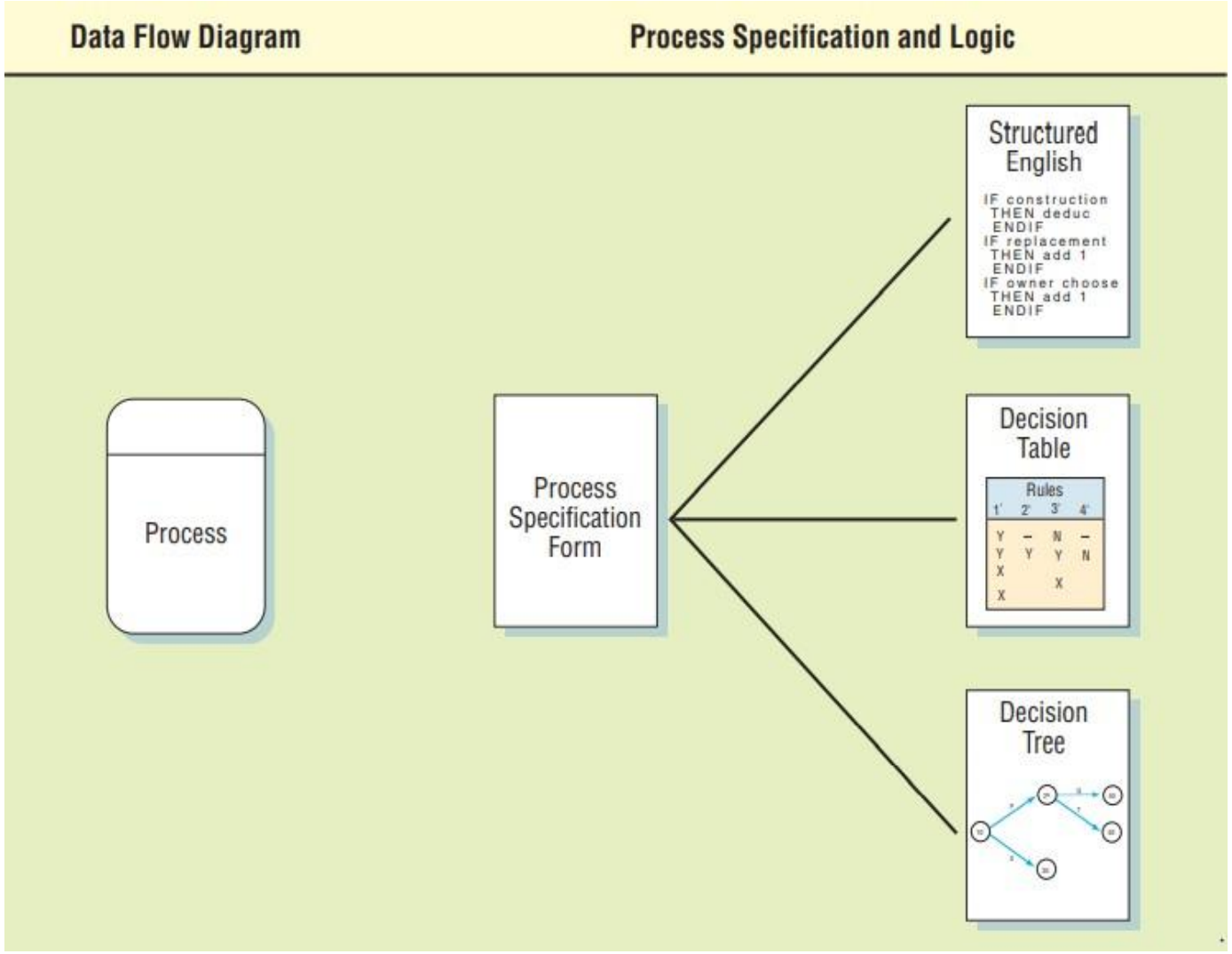
client_id	client_name	Password	Contact_no	email
1	Raju	12345	777 777 777	raju@email.com
2	Rohan	Abcdef	666 666 666	rohan@email.com
3	Sohan	54321	555 555 555	sohan@email.com
4	Mohan	14145	222 222 222	mohan@email.com

Field Name	Data type	Field Length	Constrains	Description
client_id	number	10	primary key	Client id, Auto generated
client_name	varchar	20	not null	Name of client
password	varchar	30	not null	Login password
contact_no	number	10	not null	Contact of client
email	varchar	40	not null	Client email

PROCESS SPECIFICATIONS AND STRUCTURED DECISIONS

- Process specifications - sometimes called minispecs, because they are a small portion of the total project specifications.
- Purpose - It is created for primitive processes on a data flow diagram as well as for some higher-level processes that explode to a child diagram.
- Process specifications link the process to the data flow diagram. Each process specification should be entered on a separate form or into a CASE tool screen such as the one used for Visible Analyst .

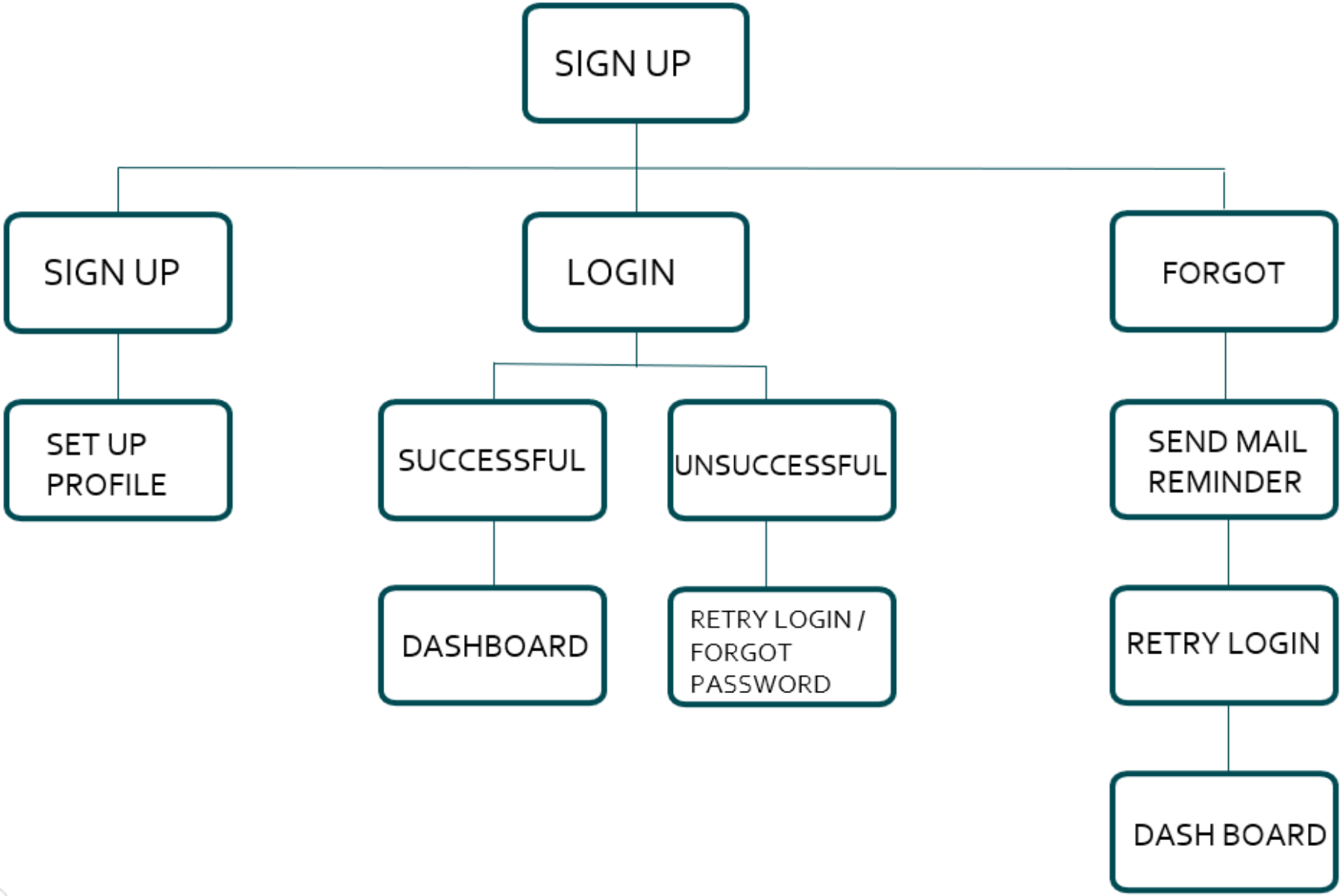
PROCESS SPECIFICATIONS AND STRUCTURED DECISIONS



HIPO CHARTS

- Hierarchical Input Process Output used for representing the modules of a system as a hierarchy and for documenting each module.
- The HIPO model was developed by IBM in the year 1970.
- HIPO diagram represents the hierarchy of modules in the software system.
- Analyst uses HIPO diagram to obtain a high-level view of system functions.
- It decomposes functions into sub-functions hierarchically. It depicts the functions performed by the system.
- HIPO diagrams are good for documentation purposes. Their graphical representation makes it easier for designers and managers to get the pictorial idea of the system structure.

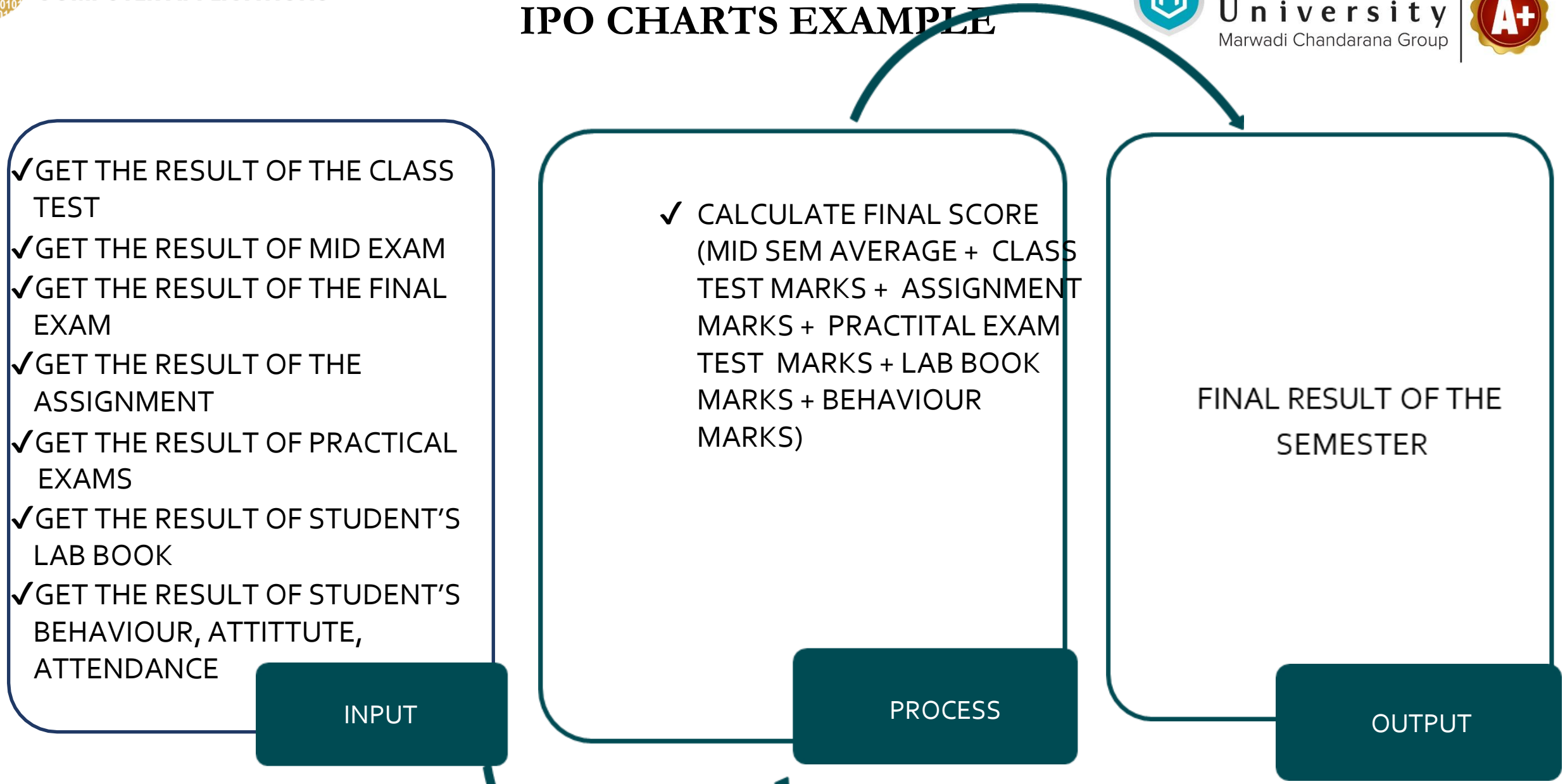
HIPO CHART EXAMPLE



IPO CHARTS

- IPO means Input Process Output diagram.
- It is very high-level diagram used for systems analysis that visually describes key inputs, business processes, resulting output with the description of each component in word.
- Purpose : An IPO Chart is one of several tools that programmers use when designing a program (before coding it). Allows you to plan out what your program needs to do.

IPO CHARTS EXAMPLE



DECISION TABLE

- It is a tabular representation of decision points and corresponding actions in a system,
- **Purpose – It is used for both complex software testing and requirements management. A decision table helps to check all possible combinations of conditions for testing and testers can also identify missed conditions easily.**
- The conditions are indicated as True(T) and False(F) values.
- While creating decision tables, you need to identify 2 things :

1. Condition
2. Actions

DECISION TABLE EXAMPLE - 1

Example 1: Make Decision Base Table for Login Screen

Conditions and Actions :

1. Username and password both are wrong. The user is shown an error message.
2. Username is correct, but the password is wrong. The user is shown an error message.
3. Username is wrong, but the password is correct. The user is shown an error message.
4. Username and password both are correct, and the user navigated to the homepage.

HOW TO MAKE DECISION BASE TABLE FOR LOGIN SCREEN

Example 1: How to make a Decision Base Table for the Login Screen

Conditions	Rule1	Rule2	Rule3	Rule4
Username (T/F)	F	T	F	T
Password (T/F)	F	F	T	T

Actions	1	2	3	4
Login Not Successful	✓	✓	✓	
Login Successful				✓

HOW TO MAKE DECISION BASE TABLE FOR LOGIN SCREEN

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8	Rule9
Conditions	TC01	TC02	TC03	TC04	TC05	TC06	TC07	TC08	TC09
UserID	Blank	Valid	Invalid	Blank	Blank	Invalid	Valid	Invalid	Valid
Password	Blank	Blank	Blank	Valid	Invalid	Valid	Invalid	Invalid	Valid
Actions									
Login Successfully									Execute
Error showing 'Invalid Credentials'	Execute	Execute	Execute	Execute	Execute	Execute	Execute	Execute	

STRUCTURED ENGLISH

- The analyst uses tools such as Structured English. It is nothing but a description of what is required to code and how to code it. Structured English helps the programmer write error-free code.

- Other methods, which use graphs or diagrams, may sometimes be interpreted differently by different people. Here, both Structured English and Pseudo-Code try to mitigate (make ease) that understanding gap

- Structured English uses plain English words in structured programming. It is not the final code but a kind of description of what is required to code and how to code it.

STRUCTURED ENGLISH

- Most programmers are unaware of the large picture of software, so they rely on what their managers tell them to do.
- Higher software management's responsibility is to provide accurate information to the programmers to develop accurate yet fast code.
- Some methods, which use graphs or diagrams, may sometimes be interpreted differently by different people.

STRUCTURED ENGLISH CONVENTIONS

1) Express all logic in terms of one of these four types: sequential structures, decision structures, case structures, or iterations (see the illustration below for examples).

1) Use and capitalize accepted keywords such as IF, THEN, ELSE, DO, DO WHILE, DO UNTIL, and PERFORM.

1) Indent blocks of statements to show their hierarchy (nesting) clearly.

1) When words or phrases have been defined in a data dictionary, underline those words or phrases to signify that they have a specialized, reserved meaning.

1) Be careful when using “and” and “or,” and avoid confusion when distinguishing between “greater than” and “greater than or equal to” and like relationships. “A and B” mean both A and B; “A or B” means either A or B, but not both.

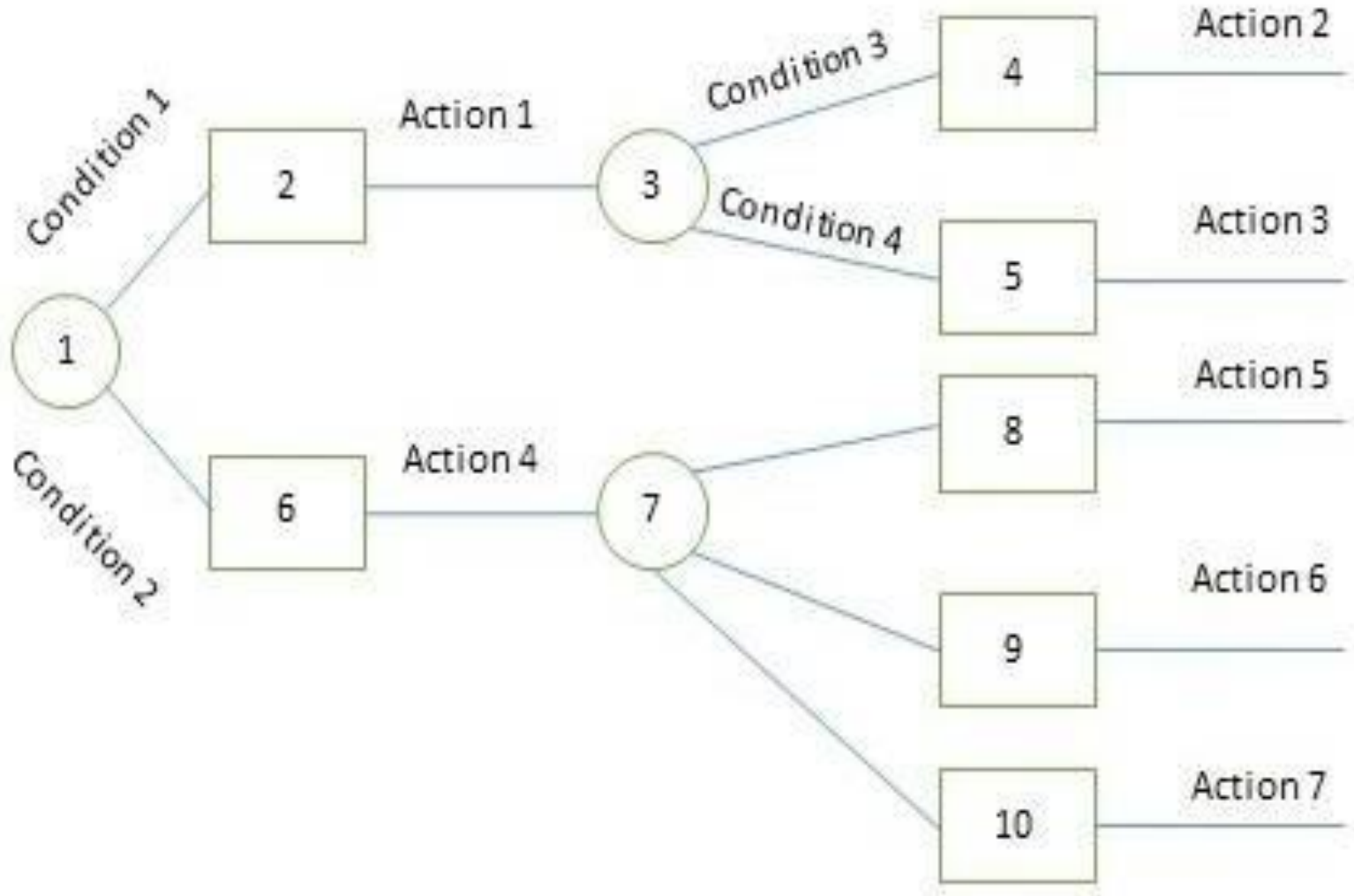
STRUCTURED ENGLISH EXAMPLE

```
IF a customer pays an advance
        THEN Give a 5% Discount
ELSE
        IF the purchase amount >=10,000
            THEN
                IF the customer is a regular customer
                        THEN Give a 5% Discount
                ELSE No Discount
                END IF
            ELSE No Discount
        END IF
END IF
```

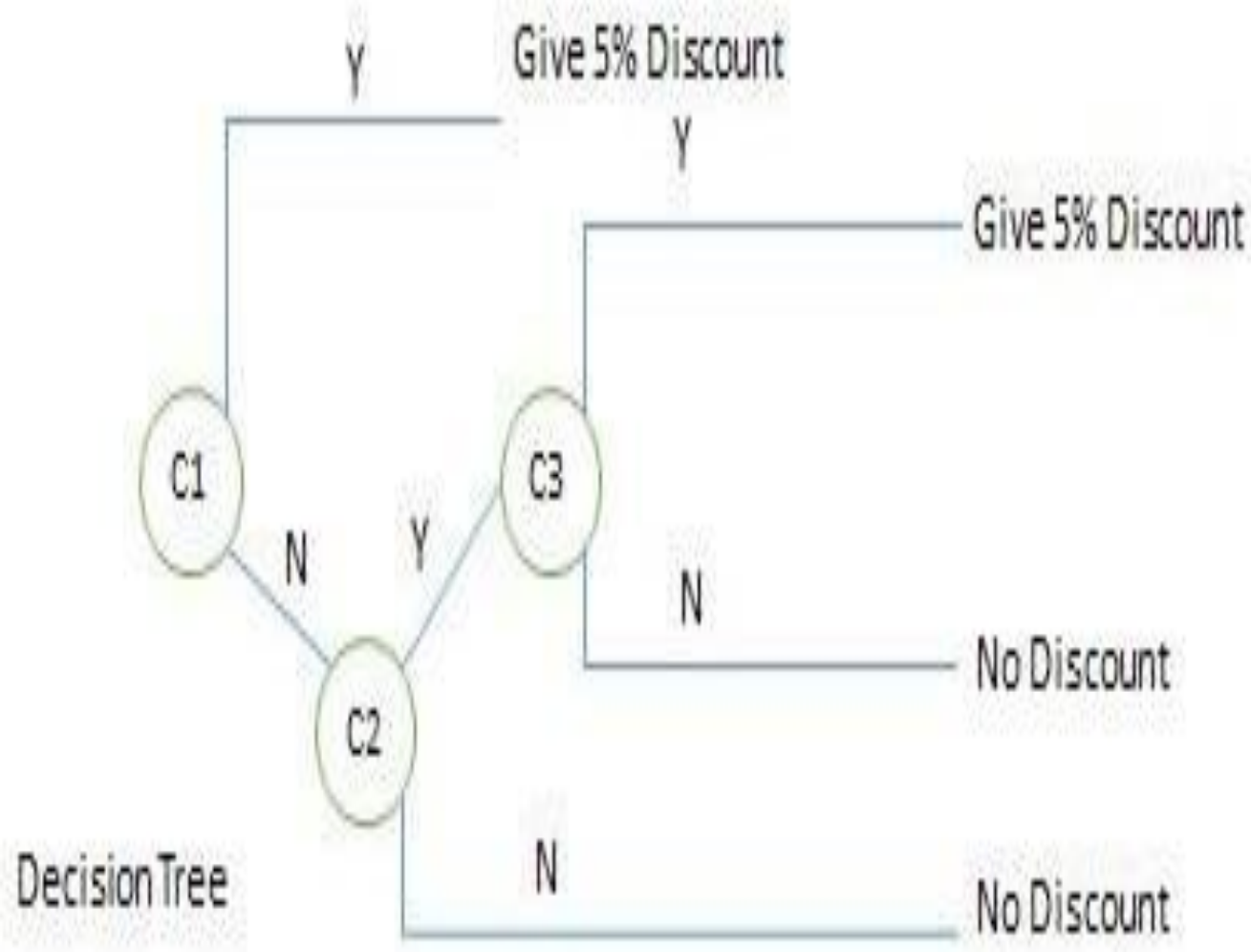
DECISION TREE

- Decision trees are used when complex branching occurs in a structured decision process.
- Trees are also useful when it is essential to keep a string of decisions in a particular sequence.

DECISION TREE



DECISION TREE



Decision Tree

Y = Yes

N = No

C1 : Advance Payment Mode

C2 : Purchase Amount >= 10,000

C3 : Regular Customer

THANK YOU

